

Why does a third of them not respond?

How this works. Groups of 5. **Leave the building.** You have **20 minutes**. Nobody expects a complete answer.

You have just started as the data person in a research group at a university hospital. The group works with the gastroenterology department on **inflammatory bowel disease**.

Patients with moderate-to-severe ulcerative colitis are started on an **anti-TNF antibody** (infliximab). It is an effective, expensive, standard-of-care drug — but roughly **35% of patients never respond at all**. Those patients spend 14 weeks on a drug that does nothing for them, get sicker, and only then move to a different therapy.

The group's ambition: **find out, before treatment starts, who will not respond.**

WHAT YOU HAVE

A biobank built over four years from **140 patients** who all started infliximab, all with a recorded outcome at week 14 (endoscopic remission: yes / no). For every patient, collected *before the first dose*:

- serum, frozen at -80°C
- whole blood in PAXgene tubes (RNA-stabilised)
- colon biopsies — one from inflamed tissue, one from adjacent non-inflamed tissue
- a stool sample
- the full clinical record: age, sex, disease duration, disease extent, previous and concomitant medication (steroids, azathioprine), smoking status, baseline CRP and faecal calprotectin

WHAT YOU HAVE TO WORK WITH

About **€50,000** for measurements, **18 months**, one PhD student and half a bioinformatician. The samples cannot be regenerated.

DISCUSS

1. What is the question, precisely? Is *who will respond* the same problem as *why do they not respond*?
2. Which material would you measure, and why that one rather than the others?
3. Which technology or technologies — and does it have to be omics at all?
4. How many of the 140, and split how?
5. What would you have at the end, and what would it justify?

It worked at fifty litres.

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You are in the process science group of a mid-sized industrial biotech. The company sells an **enzyme secreted by an engineered *Bacillus subtilis* strain**, used in detergent formulation.

- At **50 L pilot scale**: titre **4.2 g/L**, reproducible, batch after batch.
- At **10 m³ production scale**: titre **2.6–3.4 g/L**, batch-to-batch CV **18%**.
- In two of the last six production batches, the product also showed **partial proteolytic degradation** — clipped fragments on the gel, and reduced specific activity per gram.

Same strain, same seed train, same medium recipe, same 120-hour fed-batch profile.

WHAT YOU HAVE

- **Online process data from every batch for the past year** (about 40 batches): dissolved oxygen, pH, temperature, off-gas O₂/CO₂, feed rate, agitation, base addition, pressure.
- **Archived frozen broth samples** — 8 time points from every one of those batches.
- QC release data for every batch: titre, specific activity, purity.
- Raw-material lot numbers for every batch.
- The ability to run new experiments at **1–2 L bench scale** and **50 L pilot scale**.

CONSTRAINTS

You must **recommend an action in 10 weeks**. A full production campaign costs about **€250,000**, so a wrong recommendation is expensive and visible. And: **any change to the strain requires re-registration and re-validation** — that is a two-year path. A change to the *process* can be deployed in the next campaign.

DISCUSS

1. What are the plausible explanations? Rank them by how likely they are, and by how fast you could test them.
2. What would you measure — and what would you measure *first*, before spending anything?
3. Which samples from the archive, how many, and matched how?
4. You have 40 batches of data you have already paid for. What can you get out of that before you run a single new assay?
5. What is a ten-week answer, and what is a two-year answer?